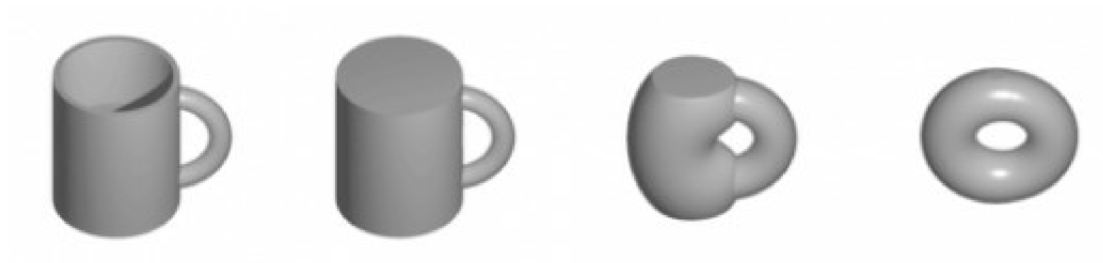
# MATH428-26S2 Topology

## Contents

## What is topology?

Roughly speaking, topology is the study of properties of spaces that remain invariant under continuous transformations, along with the properties of continuous functions between such spaces. Topology consists of four main (interlinked) streams: *point-set topology, geometric topology, algebraic topology*, and *differential topology*. We will study aspects of the first three (and in this order).



A tea cup morphs into a doughnut

You should be familiar with the notion of (countably and uncountably) infinite sets, as well as some basic real analysis (limits, continuity etc.) and the algebraic notion of a group (including subgroups, cyclic groups, isomorphisms, homomorphisms).

**Credits:** These notes are based largely on the following texts:

- Basic Topology, M. A. Armstrong, Springer 1978.
- Algebraic Topology: A First Course, M.J. Greenberg and J.R. Harper, Perseus Publishing, 1981.

# 1 Metric spaces

**Definition 1.1 (Metric and metric space)**
A *metric space* is a pair $(X, d)$ where $X$ is a set and $d$ is a function from $X \times X \to \mathbb{R}$, called a *metric* (or *distance function*), satisfying the following properties:

(M1) $d(x, y) = 0 \;\Leftrightarrow\; x = y$,

(M2) $d(x, y) = d(y, x)$, (symmetry)

(M3) $d(x, z) \leq d(x, y) + d(y, z)$, (triangle law)

where M1-M3 hold for all $x, y, z \in X$.

**Example 1.2**

1. For any set $X$, let $d(x, y) = 0$ if $x = y$ and $d(x, y) = 1$ otherwise. This metric is called the *discrete metric*.

2. Consider $(\mathbb{R}^n, d)$ where $d$ is the 'Euclidean metric' defined by:

$$d(x,y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

or the $l_1$ metric defined by

$$d(x,y) = \sum_{i=1}^{n}|x_i - y_i|$$

or the $l_\infty$ metric defined by

$$d(x,y) = \max_i |x_i - y_i|$$

3. Let $X$ be the set of continuous functions $f : [0,1] \to \mathbb{R}$ with the metric

$$d(f,g) = \sup_{x\in[0,1]} |f(x) - g(x)|.$$

**Definition 1.3 (Ball and open set)**

Let $(X, d)$ be a metric space.

- For a point $x \in X$, a *ball* $B$ of radius $r > 0$ around $x$ is the set
$$B = B(x, r) = \{y \in X \mid d(x, y) < r\}.$$
- A subset $O \subseteq X$ is *open* if for every point $x \in O$, there is a ball around $x$ that is entirely contained in $O$.

## Example 1.4

1. Every ball in a metric space is an open set.
2. The open interval $(0, \infty)$ is an open subset of $\mathbb{R}$ where $\mathbb{R}$ is equipped with the Euclidean distance.

## Remark 1.5

1. A subset $O$ of a metric space $X$ is open if and only if it is a union of balls.

Suppose $\mathcal{B}$ is a set of balls and $O = \bigcup \mathcal{B}$
Then $\forall x \in O, \exists B \in \mathcal{B}$ with $x \in B \subseteq O$
Conversely, if $O$ is open then

$O = \bigcup \{B_x, x \in O\}$ where $B_x \subseteq O$ is any open ball containing $x$.

2. Suppose that $(X, d)$ is a metric space and $x, y \in X$ are distinct. Then there exist open sets $O_x$ and $O_y$ with $x \in O_x, y \in O_y$ such that $O_x \cap O_y = \emptyset$.

Take $O_x = B(x, \delta/2)$ $O_y = B(y, \delta/2)$. then $O_x \cap O_y = \phi$.

**Definition 1.6 (Limit point and closed set)**
Let $(X, d)$ be a metric space.

- A point $p$ is a *limit point* for a set $A$ if every open set containing $p$ contains at least one point of $A \setminus \{p\}$.
- A subset $C$ of $X$ is a *closed set* if and only if it contains all of its limit points.

**Example 1.7** The limit points of the open interval $(0, 1)$ are the points in $[0, 1]$. The closed interval $[0, 1]$ is a closed subset of $\mathbb{R}$, $(0, 1]$ is neither open nor closed in $\mathbb{R}$ where $\mathbb{R}$ is equipped with the Euclidean distance.

**Remark 1.8** A subset $O \subseteq X$ is open if and only if $X \setminus O$ is closed, and $C$ is closed if and only if $X \setminus C$ is open.

If $O$ is open, let $A = X \setminus O$ and $p$ be a limit point. Suppose $p \notin A$. Then $p \in X \setminus A = O$. Since $A$ is open $\exists \delta > 0 : B(p,\delta) \subseteq O$. But since $B(p,\delta)$ is an open set containing $p$ but with no point of $A \setminus \{p\}$ $(A = X \setminus O)$. This contradicts the assumption that $p$ is a limit point of $A$, so $p \in A$. This holds $\forall$ limit points of $A$ so $A$ is closed.

**Definition 1.9 (Continuity)**
Given metric spaces $(X,d)$ and $(Y,d')$, a function $f : X \to Y$ is *continuous* if for all $x \in X$ and every $\varepsilon > 0$, there exists $\delta > 0$ so that

$$d(x,x') < \delta \Rightarrow d'(f(x), f(x')) < \varepsilon.$$

**Example 1.10**

Converse of remark 1.8.

Conversely suppose that $A = X \setminus O$ is closed and $x \in O$

Since $x$ is not a limit point of $A$, this is some open ball $B(x, \delta)$ that does not intersect $A$, so $B(x, \delta) \subseteq O$. This holds $\forall\, x \in O$

So $O$ is open.

1. Consider any function $f : X \to Y$ for which there is a $c \geq 0$ such that
$$d'(f(x), f(x')) \leq c \cdot d(x, x')$$
for all $x, x' \in X$. Then $f$ is continuous (why?).

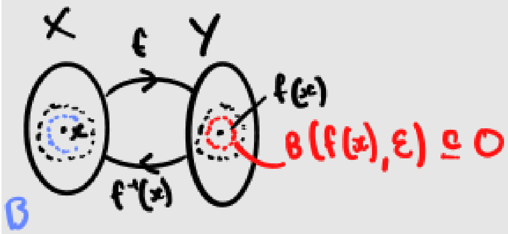
2. If $x_0 \in X$, then the function from $X$ to $\mathbb{R}$ defined by $f(x) = d(x, x_0)$ is continuous.

$$\underbrace{d(f(x), f(x'))}_{<\varepsilon} = |f(x) - f(x')| = |d(x, x_0) - d(x', x_0)| \leq \underbrace{d(x, x')}_{\delta = \varepsilon}$$

**Theorem 1.11** Given metric spaces $(X, d)$ and $(Y, d')$ a function $f : X \to Y$ is continuous if and only if for every open set $O$ in $Y$, $f^{-1}(O)$ is an open set in $X$.

$\Longrightarrow$

Suppose $f$ is cont and $O$ is an open set in $Y$ and $x \in f^{-1}(O)$




$\exists\, \varepsilon > 0 : B(f(x), \varepsilon) \subseteq O$. By continuity of $f$ $\exists\, \delta > 0$ :

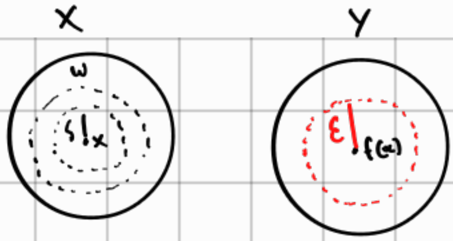
$$x' \in B(x,\delta) \Longrightarrow f(x') \in B(f(x),\varepsilon) \subseteq O$$

Thus $B(x,\delta) \subseteq f^{-1}(O)$. Since this holds $\forall\, x \in f^{-1}(O)$ this set is open.

**Remark 1.12** Two metric spaces $(X, d)$ and $(Y, d')$ are considered to be equivalent if there is an *isometry* between them; that is, a one-to-one, onto function for which $d'(f(x), f(x')) = d(x, x')$ for all $x, x' \in X$.

This is equivalent to the existence of 'non-expansive' functions $f$ from $X$ to $Y$ (that is, $d'(f(x), f(x')) \leq d(x, x')$ for all $x, x' \in X$) and $g$ from $Y$ to $X$ so that $g \circ f = 1_X$ and $f \circ g = 1_Y$ where $1_X$ and $1_Y$ are the identity

Thm 1.11 Converse.




Conversely, suppose $f$ has the property described. Let $x \in X$, and $\varepsilon > 0$. Then open ball $B(f(x), x)$ is an open set in $Y$.

So $f^{-1}(B(f(x), \varepsilon))$ is an open set $(W)$ in $X$ with $x \in W$, so $\exists \delta > 0 : B(x, \delta) \subseteq W$.

This implies that $\forall y : d_X(x,y) \leq \delta$
$$\implies d_Y(f(x), f(y)) \leq \varepsilon$$

This holds $\forall \varepsilon > 0$ $f$ is continuous.

$f^{-1}(O)$ is open in $X$ $\forall$ Open sets in $Y$

functions on $X$ and $Y$, respectively.

**Quiz** For the Euclidean metric on $\mathbb{R}^2$ or $\mathbb{R}^3$, what sorts of functions are isometries?

**Interlude 1.13** Before proceeding to the next section we summarise some basic properties of sets. Given sets $A, B$ with $A \subseteq B$, we let $B \setminus A$ be the complement of $A$ in $B$. Given a collection $\mathcal{C} = \{A_i \mid i \in I\}$ of subsets of a set $X$ (note $I$ might be uncountably infinite), recall De Morgan's laws for the complements of unions and intersections.

$$X \setminus \bigcup_{i \in I} A_i = \bigcap_{i \in I} (X \setminus A_i), \text{ and } X \setminus \bigcap_{i \in I} A_i = \bigcup_{i \in I} (X \setminus A_i)$$

# 2 Topological spaces

**Definition 2.1 (Topology)**
A *topological space* is a pair $(X, \tau)$ where $X$ is a set and $\tau$ is a set of subsets of $X$ (called a *topology*) satisfying the following properties.

(O1) The empty set $\emptyset$ and the set $X$ are both in $\tau$.

(O2) The union of any collection of sets in $\tau$ is contained in $\tau$.

(O3) The intersection of any finite collection of sets in $\tau$ is also contained in $\tau$.

The elements of $\tau$ are called *open sets*. We will often just refer to $(X, \tau)$ as 'the (topological) space $X$'.
A set $C$ is said to be *closed* if $X \setminus C$ is open.

**Example 2.2**

1. $(0, 1)$ is an open subset of the real line $\mathbb{R}$ and $[0, 1]$ is a closed set (since it is the complement of $(-\infty, 0) \cup (1, \infty)$. The half-open interval $(0, 1]$

is neither open nor closed. In $\mathbb{R}$, the sets $\emptyset$ and $\mathbb{R}$ are both open and closed (!), but these are the only such sets in $\mathbb{R}$. (In some topological spaces, sets other than $\emptyset$ and the full set can be both open and closed ('clopen').)

2. Let $(X, d)$ be a metric space. The collection $\tau$ of all open sets (as defined earlier in terms of balls, see Definition 1.3) forms a topology (why?). Often two different metrics will generate the same topology (e.g., this holds for the three metrics on $\mathbb{R}^n$ discussed in Example 1.2.2).

3. Let $X$ be a set with the topology $\tau$ equal to the set of **all** subsets of $X$. This is called the *discrete topology* on $X$. Note that in this topology every open set is also closed. This is a special case of a metric space topology since it arises from the metric $d$ on $X$ for which $d(x, y) = 0$ if $x = y$ and $d(x, y) = 1$ otherwise.

**Remark 2.3** Finite unions of closed sets are closed, and finite or infinite intersections of closed sets are closed.

Thinking of $\mathbb{R}$ as a metric space under the Euclidean distance, open intervals $(a, b)$ are open sets, as are unions of them. In fact, we have the following result:

**Theorem 2.4** Any non-empty open set in $\mathbb{R}$ or in $(0, 1)$ is a countable disjoint union of non-empty open intervals.

Note, however, that a closed set in $\mathbb{R}$ is not necessarily a countable disjoint union of closed intervals. Indeed, there is a famous closed subset of $[0, 1]$ (a 'Cantor space', see Example 2.12) that is uncountable and that contains no intervals of the form $[a, b]$ where $a < b$.

# Thm 2.4 Proof

Define a relation $\sim$ on the non-empty open set $O$, by $x \sim y$ if the closed interval w/ endpoints $x, y$ is contained in $O$

Then $\sim$ is an equivalence relation. $\begin{cases} x \sim x \\ x \sim y \Rightarrow y \sim x \\ x \sim y \wedge y \sim z \Rightarrow x \sim z \end{cases}$

And so it partitions the real line into non-empty intervals in $\mathbb{R}$, and since $O$ is open, they are non empty open intervals in $O$.

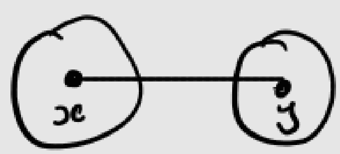
The number of classes is countable since each non-empty open interval $I$ contains a rational # $r(I)$, so the function $I \to r(I)$ is a 1-to-1 function from the equivalence classes in a subset of $\mathbb{Q}$

**Quiz** Is the subset of rationals or irrationals an open subset of $\mathbb{R}$, a closed subset of $\mathbb{R}$?

**Example 2.5 (The co-finite topology on $\mathbb{R}$)**
Take $X$ to be the set of all real numbers with the so-called *finite complement topology* (or 'co-finite topology'). Here, a set is open if its complement is either finite or all of $X$ (i.e. the open sets are $\emptyset, X$ and sets of the form $\mathbb{R} \setminus \{x_1, \ldots, x_m\}$ for some $m \geq 1$).
For example, for each $p \in \mathbb{R}$, the singleton set $\{p\}$ is a closed set in this topology.

**Remark 2.6** The co-finite topology on $\mathbb{R}$ cannot be realized by a metric, i.e., there is no metric on $\mathbb{R}$ so that the open sets in this metric space equal the open sets in the co-finite topology.

$x, y$
$x \neq y$




find

$O_x \cup O_y = \varphi$

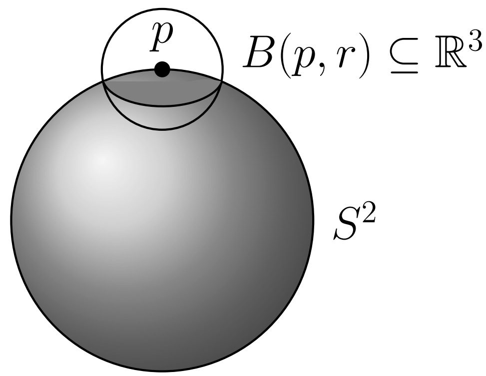
**Example 2.7 (Zariski topology on $\mathbb{C}^n$)**
Given a collection $P$ of polynomials in $x_1, \ldots, x_n$ with complex coefficients, the closed sets are all sets of the form $\{x \in \mathbb{C}^n \mid p(x) = 0, \text{for all } p \in P\}$ (together with $\emptyset$ and the full set $\mathbb{C}^n$).

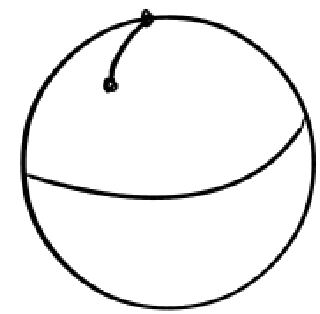
**Example 2.8 (Finite topologies)**
Let $P$ be a finite set with a preorder $\leq$ (that is, $\leq$ is reflexive and transitive). A *downset* is a subset $P'$ of $P$ that satisfies the following property: If $x \in P'$ and $y \leq x$ then $y \in P'$. Let $dP$ denote the collection of downsets of $P$ (including $\emptyset$ and $P$). Then $(P, dP)$ is a topological space. Every topology on a finite set can be obtained in this way from a preorder. For the particular case where $P = \{a, b, c, d\}$ with $a < c, a < d$ and $b < c, b < d$ and with no other inequalities, this space is called the *pseudocircle*.

**Definition 2.9 (Induced subspace topology)**
If $Y \subseteq X$ and $\tau_X$ is a topology on $X$, we define the *induced subspace topology* on $Y$ as $\tau_Y = \{O \cap Y \mid O \in \tau_X\}$.

**Example 2.10** Consider the sphere

$$S^2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1^2 + x_2^2 + x_3^2 = 1\}.$$



Induced topology on $S^2$

This sits inside $\mathbb{R}^3$ and so the usual topology on $\mathbb{R}^3$ (induced for example by any of the usual metrics on $\mathbb{R}^3$) induces a topology on $S^2$. A ball in the Euclidean metric cuts $S^2$ in an open 'curved disc' embedded in $S^2$. This topology is the same as that induced by the intrinsic *geodesic* metric on $S^2$ (where the distance between two points is the length of the shortest path on $S^2$ from one point to the other, that is, the length of the shortest arc on a great circle through the two points).

## Remark 2.11

1. Note that an open set $O$ of the subspace $Y$ need not be an open set of the larger space $X$. For example, if $Y = [0, 2)$ and $X = \mathbb{R}$, then $[0, 1)$ is an open set of $Y$ but not of $X$.

   However if $Y$ is an open set in $X$ (i.e. $Y \in \tau_X$), then an open set of $Y$ is an open set of $X$ (why?).

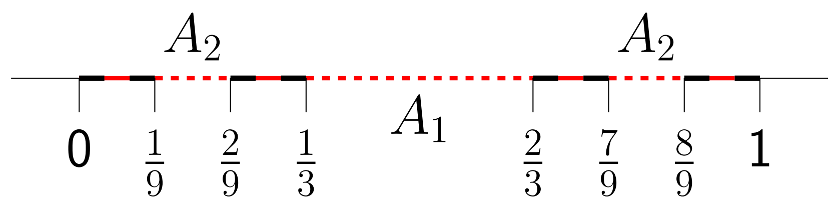
2. A subset $A \subseteq Y$ is closed in $\tau_Y$ if and only if $A = B \cap Y$ for some closed set $B \subseteq X$.

   Similarly as for open sets, a closed set $A$ of the subspace $Y$ need not be a closed set of the larger space $X$. However, if $Y$ is a closed set in $X$ (i.e. $X \setminus Y \in \tau_X$), then a closed set of $Y$ is a closed set of $X$ (why?).

## Example 2.12 (Induced topology: Cantor space)

Figure 1: Georg Cantor c. 1870

Consider the subspace $\mathcal{C}$ of $[0,1]$ (with the usual topology) obtained by removing the set $A_1$ consisting of the open interval $(\frac{1}{3}, \frac{2}{3})$, then removing the set $A_2$ consisting of the (open) middle-thirds of the two resulting subintervals, and continuing in this fashion.




Thus

$$\mathcal{C} = \left\{ x \in [0,1] \;\middle|\; x = \sum_{n \geq 1} \frac{2x_n}{3^n}, x_n \in \{0,1\} \right\}.$$

Note that $\mathcal{C}$ is non-empty (indeed infinite) and is an important topological space (called the 'Cantor set' or 'Cantor space'). We will consider it further later.
The Cantor space is a closed set in $[0,1]$.

$$C = [0,1] \setminus \left( \bigcup_{n=1}^{\infty} \left( \bigcup_{k=0}^{3^{n-1}-1} \left( \frac{3k+1}{3^n}, \frac{3k+2}{3^n} \right) \right) \right)$$

**The Devil's staircase**

Using Cantor space one can construct a continuous real function $f : [0,1] \to [0,1]$ that is 'almost everywhere flat' and yet $f(0) = 0$ and $f(1) = 1$.

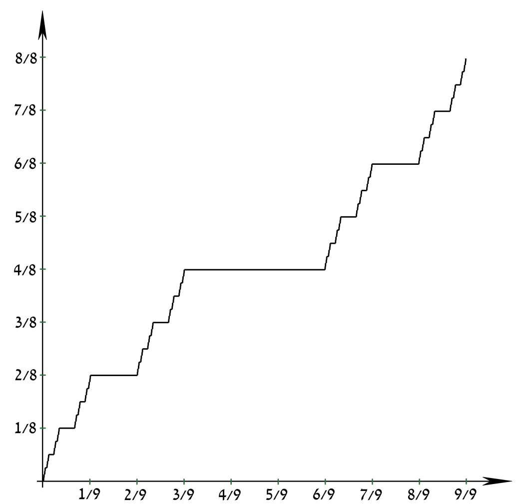



Figure 2: Devil's starcase

Two subspaces of $\mathbb{R}^n$ that play an important role in this course are the following.

n dim closed ball

$E^0$ point
$E^1$ line $S^0$
$E^2$ $S^1$
$S^2$

**Definition 2.13 ($n$-cell and sphere)**
The (closed) $n$-cell in $\mathbb{R}^n$ is

$$E^n := \{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n \mid x_1^2 + \cdots + x_n^2 \leq 1\}.$$

Its boundary is the *sphere* of dimension $n-1$:

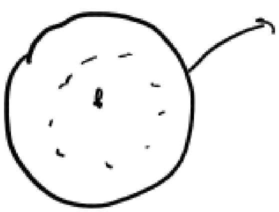
$S^0$ is boundary of $E^1$

$$S^{n-1} := \{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n \mid x_1^2 + \cdots + x_n^2 = 1\}.$$

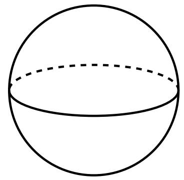
For example, $E^2$ is just the disc (in the plane, centred on the origin) while $S^1$ is its boundary circle. Similarly, $E^3$ is a ball and $S^2$ its boundary sphere. Note that $S^0$ consists of just two isolated points $1$ and $-1$ and $E^0$ consists of a single point, while $E^1$ is the closed interval $[-1, 1]$.
Other subspaces of $\mathbb{R}^3$ (and of $\mathbb{R}^4$) are shown in the figure below.

**Definition 2.14** Let $(X, \tau)$ be a topological space.

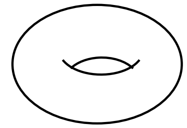
N can be closed



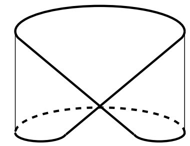
- Given a point $x$ of $X$, we call a subset $N$ of $X$ a *neighbourhood* of $x$ if there is an open set $O$ (i.e. $O \in \tau$) such that $x \in O \subseteq N$.
- The *interior* of a set $A$, written $\mathrm{int}(A)$ or $\mathring{A}$, is the union of all open sets contained within $A$.



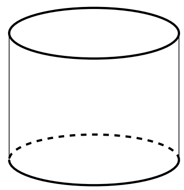
(a) Sphere



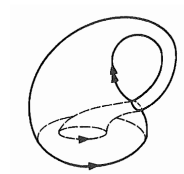
(b) Torus



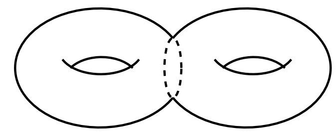
(c) Möbius strip



(d) Cylinder



(e) Klein bottle



(f) Double torus

Figure 3: Some simple topological spaces. Examples (a, b, c, d, f) are subspaces of $\mathbb{R}^3$, while the Klein bottle (e) (defined in 5.11) can only be realized as a subset of $\mathbb{R}^4$

# Remark 2.15

1. The interior $\mathring{A}$ is an open set, and is the largest open subset in $X$ that is entirely contained in $A$. [why?]

Interior is a union of open sets

## Example 2.16

1. Each half-open interval $[0, t)$, $0 < t \leq 1$, is a neighbourhood of 0 in $[0, 1]$.

2. The open interval $(0, 1)$ is the interior of the subset $[0, 1]$ of $\mathbb{R}$.
   Largest open set contained in $[0,1]$

3. The *open* $n$*-cell* is the interior of the closed $n$-cell:

$$\mathring{E}^n := E^n \setminus S^{n-1} = \{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n \mid x_1^2 + \cdots + x_n^2 < 1\}.$$

4. For the Cantor space $C$ as a subspace of $\mathbb{R}$, what is $\mathring{C}$?
   $\emptyset$

**Remark 2.17** The function $A \mapsto \operatorname{int}(A)$ satisfies the following four properties of a 'topological ~~closure~~ operator':
interior

- $\operatorname{int}(A) \subseteq A$, (intensive)
- $A \subseteq B \Rightarrow \operatorname{int}(A) \subseteq \operatorname{int}(B)$, (monotone)
- $\operatorname{int}(\operatorname{int}(A)) = \operatorname{int}(A)$, (idempotent)
- $\operatorname{int}(A \cap B) = \operatorname{int}(A) \cap \operatorname{int}(B)$. (topological)

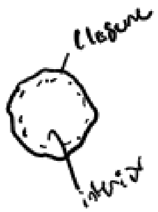
Proof of topological

$int(A) \cap int(B)$ is an intersection of two open sets contained in $A \cap B$ so $int(A) \cap int(B) \subseteq int(A \cap B)$

Also $A \cap B \subseteq A$ so $int(A \cap B) \subseteq int(A) \cap int(B)$

$\subseteq B$

**Definition 2.18** Let $(X, \tau)$ be a topological space and let $A$ be a subset of $X$. The *closure* of $A$, written $\mathrm{cl}(A)$ or $\overline{A}$, is the intersection of all closed sets in $X$ that contain $A$.



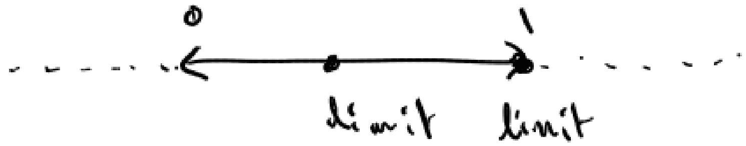

Thus, $\overline{A}$ is the smallest closed set containing $A$ [why?].

**Remark 2.19** The function $A \mapsto \mathrm{cl}(A)$ satisfies the following four properties of a 'topological closure operator':

- $A \subseteq \mathrm{cl}(A)$, (extensive)
- $A \subseteq B \Rightarrow \mathrm{cl}(A) \subseteq \mathrm{cl}(B)$, (monotone)
- $\mathrm{cl}(\mathrm{cl}(A)) = \mathrm{cl}(A)$, (idempotent)
- $\mathrm{cl}(A \cup B) = \mathrm{cl}(A) \cup \mathrm{cl}(B)$. (topological)

$\mathrm{cl}(A) \cup \mathrm{cl}(B)$ is closed as finite union of closed sets are closed

finish proof






**Definition 2.20** (Limit Point) Let $(X, \tau)$ be a topological space. A point $p$ of $X$ is a *limit point* of a subset $A$ of $X$ if every neighbourhood of $p$ contains at least one point of $A \setminus \{p\}$.

Note: 'neighborhood' can be replaced by 'open set'.

**Example:** Consider the real numbers with the finite complement topology. If $A$ is an infinite subset of $\mathbb{R}$ (say the set of all integers), then every point of $\mathbb{R}$ is a limit point of $A$. On the other hand, a finite subset of $\mathbb{R}$ has no limit points in this topology.

If $A$ is infinite and $p \in \mathbb{R}$, Let $O$ be an open set containing $p$ then $O = \mathbb{R} \setminus F$ for a finite subset $F$ of $\mathbb{R}$

thus $O$ contains at least one point of $A \setminus \{p\}$. This helps $\forall\, O$ so $p$ is a limit point of $A$

**Theorem 2.21** A set is closed if and only if it contains all its limit points.

If $A$ is finite, either

(i) $p \in A$ or

(ii) $p \notin A$

In case (i), $\mathbb{R} \setminus (A \setminus \{p\})$ is a open set containing $p$ that has empty intersection w/ $A \setminus \{p\}$ so $p$ is not a limit point of $A$.

In case (ii), $\mathbb{R} \setminus A$ is an open set cont $p$ that has empty intersection w/ $A \setminus \{p\}$. So $p$ is not a limit point of $A$.

If $A$ is closed, then $X\setminus A$ is open. No point of $X\setminus A$ can be a limit point of $A$ since if $p \in X\setminus A$ then $O = X\setminus A$ is an open set cont $p$ w/ $O \cap (A\setminus\{p\}) = \emptyset$. Thus $A$ cont all limit points

Conversely say $A$ contains all its limit points. Let $x \in X\setminus A$. Since $x$ is not a limit point of $A$ $\exists$ open set $O_x$, with $x \in O_x$ which has $A \cap O_x = \emptyset$ this holds $\forall x \in X\setminus A$. So $X\setminus A = \bigcup_{x \in X\setminus A} O_x$ is open, so $A$ is closed.

**Definition 2.22 (Dense set)**
A subset $Y$ of $X$ is *dense* if its closure equals $X$.

**Example 2.23** The rational numbers (or the irrational numbers) form a dense subset of $\mathbb{R}$.

**Definition 2.24 (Base for a topology)**
Suppose we have a topology on a set $X$ and a collection $\beta$ of open sets such that every open set is a union of members of $\beta$. Then $\beta$ is called a *base* for the topology.

**Proposition 2.25** A collection $\beta$ of open sets in a topological space $(X, \tau)$ is a base for $\tau$ if and only if for every $x \in X$ and every neighbourhood $N$ of $x$, there is always an element $B$ of $\beta$ such that $x \in B \subseteq N$.

**Example 2.26**

1. Consider the Euclidean topology of $\mathbb{R}$. The set of all open intervals $\{(a, b) \mid a < b\}$ is a base for this topology. So too is the (countable!) collection of open intervals $(a, b)$ where $a$ and $b$ are both rational (why?).

Let $B'$ be the set of open intervals $(a,b)$ w/ $a,b \in \mathbb{Q}$ then $B'$ is a base for the same topology as $B$ generates. Since for any interval $(a,b)$ $a,b \in \mathbb{R}$ select $a_n, b_n \in \mathbb{Q}$ $a_n \downarrow a$, $b_n \uparrow b$ $a_n < b_n$

Then $(a,b) = \bigcup_{n \geq 1} (a_n, b_n)$

2. Consider any metric space. The set of all open balls is a base for the associated topology.

Each open set is the union of open balls

$\forall x \in O$, let $B_x$ an open ball in $O$. Then $O = \bigcup_{x \in O} B_x$

3. If $D$ is dense in a metric space, then the set of all open balls with centres in $D$ and rational radii is a base for the associated topology.

**Theorem 2.27** Let $\beta$ be a non-empty collection of subsets of a set $X$. If the intersection of any finite number of members of $\beta$ is always in $\beta$ (or empty) and if $\bigcup \beta = X$, then $\beta$ is a base for a topology on $X$.

Take $\tau$ to be the collection of all unions of members of $\beta$ as its open sets (plus $\phi$) and check $\tau$ satisfies the axioms

(Below)

**Definition 2.28 (Product space)**
Let $(X, \tau)$ and $(Y, \tau')$ be topological spaces. The *product space* of these two spaces is the topological space $(X \times Y, \tau'')$ where

$$X \times Y = \{(x, y) \mid x \in X, y \in Y\}$$

and a base for the topology $\tau''$ is the collection of sets of the form

$$O \times O' = \{(x, y) \mid x \in O, y \in O'\}$$

where $O \in \tau$ and $O' \in \tau'$.

Or not

**Example 2.29** $\mathbb{R} \times \mathbb{R}$ equipped with the product topology is the same as $\mathbb{R}^2$ with the topology that comes from the metric $d_\infty$.

**Remark 2.30**

1. If $\beta$ is base for $\tau$ and $\beta'$ is a base for $\tau'$, then a base for the product topology $\tau''$ is the collection of sets of the form $O \times O'$ where $O \times O'$ where $O \in \beta$ and $O' \in \beta'$.

2. A set $N \subseteq X \times Y$ is a neighbourhood of $(x, y)$ in $(X \times Y, \tau'')$ if and only if there are open neighbourhoods $O_x, O_y$ of $x$ and $y$ (respectively) with $O_x \times O_y \subseteq N$.

**Interlude 2.31** For the next section, we summarise some basic properties of sets. Let $f : X \to Y$ be a function. For a subset $B$ of $Y$, the *pre-image* of $B$ under $f$, denoted $f^{-1}(B)$, is the set $\{x \mid f(x) \in B\}$. Notice that $f^{-1}(B \cup B') = f^{-1}(B) \cup f^{-1}(B')$. Similarly, if $A, A' \subseteq X$, then $f(A \cup A') = f(A) \cup f(A')$. More generally, given a collection $\mathcal{C}$ of sets, let $\cup\mathcal{C}$ be their union and $\cap\mathcal{C}$ their intersection. Let $\{A_i \mid i \in I\}$ be a collection of subsets of $X$ (note that $I$ might be uncountably infinite) and $\{B_i \mid i \in I\}$ be a collection of subsets of $Y$. Then:

$$f\left(\bigcup_{i\in I} A_i\right) = \bigcup_{i\in I} f(A_i), \qquad f\left(\bigcap_{i\in I} A_i\right) \subseteq \bigcap_{i\in I} f(A_i),$$
$$f^{-1}\left(\bigcup_{i\in I} B_i\right) = \bigcup_{i\in I} f^{-1}(B_i), \quad f^{-1}\left(\bigcap_{i\in I} B_i\right) = \bigcap_{i\in I} f^{-1}(B_i).$$

The second (inclusion) is an equality when $f$ is one-to-one.

# 3 Continuous functions, homeomorphisms and embeddings

**Definition 3.1 (Continuous function)**
Let $(X, \tau)$ and $(Y, \tau')$ be topological spaces. A function $f : X \to Y$ is *continuous* if and only if the pre-image of every open set in $Y$ is an open set in $X$ (i.e. for all $O' \in \tau', f^{-1}(O') \in \tau$).
We will often refer to a continuous function as simply a *map*.

**Example 3.2** Notice that for two topological spaces $X_1$ and $X_2$ we have two (continuous onto) 'projection' maps $p_j : X_1 \times X_2 \to X_j$ $(j = 1, 2)$. Each $p_j$ maps open sets to open sets (this is clear for the base open sets, and each open set in $X_1 \times X_2$ is a union of base sets).

**Lemma 3.3**

1. A function $f : X \to Y$ is continuous if and only if the pre-image of every closed set in $Y$ is a closed set in $X$.

2. A function $f : X \to Y$ is continuous if for all $x \in X$ and any neighbourhood $V$ of $f(x)$ there is a neighbourhood $U$ of $x$ such that $f(U) \subseteq V$.

3. A composition of two continuous functions is continuous (that is, if $f : X \to Y$ is continuous, and $g : Y \to Z$ is continuous, then $g \circ f : X \to Z$ is continuous).

Topology is full of counterintuitive and surprising results. Here is one (the existence of a 'space filling curve'), which we state without proof.

**Theorem 3.4** There is a continuous function $f : E^1 \to E^2$ that is onto. Moreover, for any $n > 1$, there is a continuous function from $E^1$ onto $E^n$ and a continuous function from $(0, 1)$ onto $\mathbb{R}^n$.

Sometimes, we have continuous functions defined in different parts of a space and we want to combine them to get a continuous function for the whole space. The following lemma allows us to do this.

**Lemma 3.5 (Glueing lemma)**
Suppose that $X = A \cup B$ where $A$ and $B$ are closed. Let $f : X \to Y$ be a function with $\alpha = f|_A$ and $\beta = f|_B$ being both continuous. Then $f$ is continuous.

**Remark 3.6** Note that the glueing lemma does not hold without the assumption that both $A$ and $B$ are closed.

**Definition 3.7 (Homeomorphism)**
Given two topological spaces $(X, \tau)$ and $(Y, \tau')$, a function $f : X \to Y$ is said to be a *homeomorphism* from $X$ to $Y$ if $f$ is a bijection that is continuous, and $f^{-1}$ is also continuous.
When such a function exists, we say that the two topological spaces are *homeomorphic*, written $X \cong Y$.

**Remark 3.8** This last condition is equivalent to requiring that $f$ maps open sets to open sets; such a map is called *open*. Equivalently, $f$ maps closed sets to closed sets; such a map is then called *closed*.
Equivalently, $(X, \tau)$ and $(Y, \tau')$, are homeomorphic if there is a bijection $f$ from $X$ to $Y$ which induces a bijection between $\tau$ and $\tau'$ (via the map

$O \mapsto f(O)$).

## Example 3.9

1. The open interval $(0,1)$ is homeomorphic to $\mathbb{R}^{>0}$ and to $\mathbb{R}$.

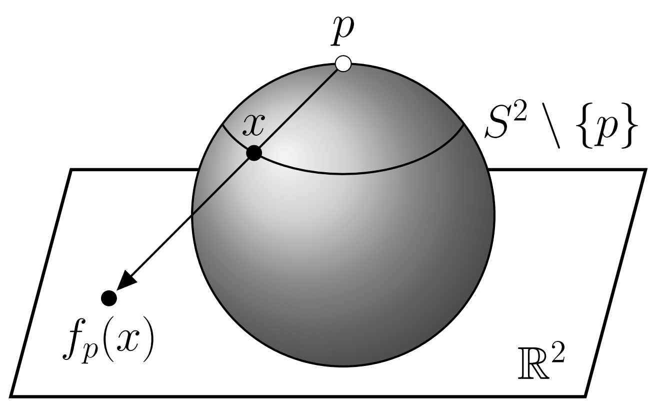
2. $S^1$ is homeomorphic to the boundary of a square.

3. The open interval $(0,1)$ and the circle minus a point $S^1 \setminus \{p\}$ are homeomorphic. What is an example of a homeomorphism?

4. Consider the half-open interval $[0,1)$ and the circle $S^1$. There is a function $f$ from $[0,1)$ onto $S^1$ which is one-to-one, onto and continuous, but its inverse is not continuous. [We will see later these two spaces are not homeomorphic.]

5. $S^n \setminus \{p\}$ is homeomorphic to $\mathbb{R}^n$ for all $n \geq 1$.

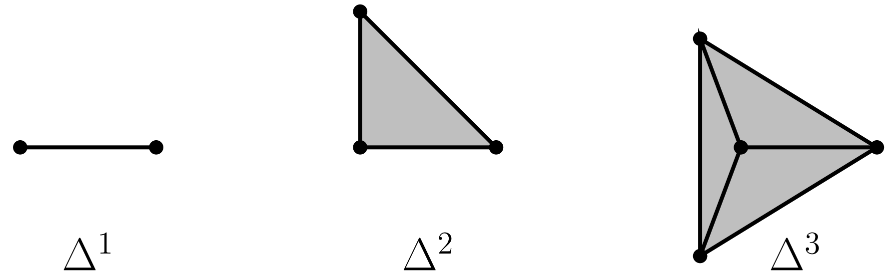
   For $n = 2$, a homeomorphism is provided by stereographic projection of $S^2 \setminus \{p\}$ from the point $p$ onto $\mathbb{R}^2$ as indicated in the figure below.




Stereographic projection

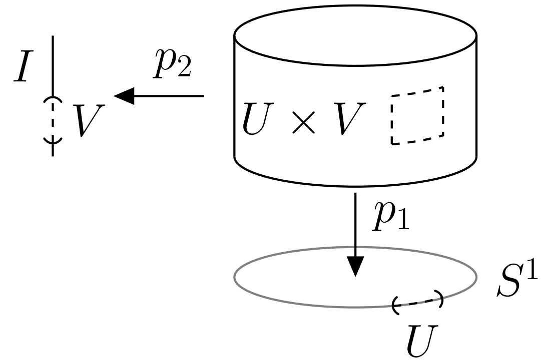
6. The closed $n$–cell $E^n$ is homeomorphic to $[0,1]^n$ and to the $n$-*simplex* $\Delta^n$ defined by:

$$\Delta^n := \{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n \mid x_1 + \cdots + x_n \leq 1, x_i \geq 0\}.$$




$1$-, $2$- and $3$-simplices

7. $\mathbb{R} \times \mathbb{R}$ is homeomorphic to $\mathbb{R}^2$.
8. The product $(0,1) \times (0,1)$ is homeomorphic to $\mathbb{R}^2$.
9. $S^1 \times [0,1]$ is homeomorphic to a cylinder, see the figure below.




Cylinder as a direct product

Homeomorphisms are to topology what isomorphisms are to group theory or isometries are in metric space theory (i.e. bijections that preserve 'structure'). Thus two topological spaces that are homeomorphic are regarded as topologically equivalent or identical.
As with isomorphisms in algebra, the composition of two homeomorphisms is a homeomorphism (and the set of homeomorphisms from a space to itself forms a group) and 'homeomorphic' is an equivalence relation on topological spaces.

The following result will be useful later.

**Lemma 3.10** A continuous map $f$ from one topological space $X$ to another space $Y$ is a homeomorphism if and only if there is a continuous map $g$ from $Y$ to $X$ so that $g \circ f = 1_X$ and $f \circ g = 1_Y$ where $1_X$ and $1_Y$ are the identity functions on $X$ and $Y$ respectively.

**Definition 3.11 (Embedding)**
If $X$ and $Y$ are topological spaces and $h$ is a homeomorphism from $X$ to a subspace $Y'$ of $Y$, then we say that $h$ is an *embedding* of $X$ into $Y$.

**Example 3.12**

1. The inclusion map $x \mapsto x$ from $S^2$ into $\mathbb{R}^3$ is an embedding of $S^2$ into $\mathbb{R}^3$.
2. $S^1 \times S^1 = \{(u, v, x, y) \in \mathbb{R}^4 \mid u^2 + v^2 = x^2 + y^2 = 1\}$ embeds into $\mathbb{R}^3$ via the map $h(u, v, x, y) = ((2 + u)x, (2 + u)y, v)$.

We will see that there is no embedding of $S^2$ into $\mathbb{R}^2$. The Klein bottle (illustrated in an Figure 3(e) and defined more precisely in Example 5.6) cannot be embedded in $\mathbb{R}^3$ but can be in $\mathbb{R}^4$.

## Infinite product spaces

The concept of a product of topological spaces extends to infinite collections of spaces. For any finite or infinite set $I$, let $((X_i, \tau_i) : i \in I)$ be any collection of topological spaces, and let $X = \prod_{i \in I} X_i$. Define a topology on $X$, by taking as a base the sets $\prod_{i \in I} O_i$ where each $O_i$ is an open set in $X_i$ and, in addition:

$O_i = X_i$, for all but finitely many values of $i \in I$.

This forms the base of a topology on $X$ [why?]. Moreover, it is the 'coarsest' topology for which the projection maps $\pi_i : X \to X_i$ are continuous.

For example, let $X_i = \mathbb{R}$ for all $i \in \mathbb{N}$. Then $\mathbb{R}^\infty = \prod_{i=1}^{\infty} \mathbb{R}$ is all (countably infinite) sequences of real numbers, and a base is the set of products, in any order, of (a finite number of) open intervals and a countably infinite number of copies of $\mathbb{R}$. The infinite dimensional sphere $S^\infty$ is the subspace of $\mathbb{R}^\infty$ defined by $\sum_{i\geq 1} x_i^2 = 1$ (it can be viewed as the infinite collection of nested spheres $S^1, S^2, \cdots$ where $S^k$ embeds as $S^{k+1}$ as its equator).

A simpler example is the countably infinite product of the 2-element discrete space $\{0,1\}$. Let $\{0,1\}^\infty = \prod_{i\geq 1}\{0,1\}$, where $\{0,1\}$ has the discrete topology. What does this space look like? It turns out to be (topologically) identical to a space we have met already!

**Example 3.13**

# 4 Three basic properties of topological spaces

## 4.1 Hausdorff property

**Definition 4.1 (Hausdorff space)**
A space $X$ is *Hausdorff* if for any two points $x, y \in X$, with $x \neq y$, there exist disjoint open sets $O_x$ and $O_y$ with $x \in O_x$ and $y \in O_y$.

We saw in the Metric Space section that every metric space is Hausdorff. Most (but not all) of the topological spaces we will consider satisfy this condition.

**Example 4.2**

1. The set of real numbers with the finite complement topology (see Example 2.5) is not Hausdorff.

2. If $X$ is Hausdorff, and $Y \subset X$ then $Y$ (with the induced subspace topology) is also Hausdorff.

3. A product of two (or more) spaces is Hausdorff if and only if each space in the product is Hausdorff.

One basic property of Hausdorff spaces is the following.

**Proposition 4.3** If $X$ is Hausdorff and $x \in X$, then $\{x\}$ is closed.

The Hausdorff properties turns out to be exactly the condition we need to prove a variety of theorems. Here is one example.

**Theorem 4.4** Suppose $f$ and $g$ are continuous functions from a space $X$ into a Hausdorff space $Y$ and suppose that $f$ and $g$ agree on a subset $X'$ of $X$. If $X'$ is dense in $X$, then $f = g$.

**Corollary 4.5** Suppose $f$ and $g$ are continuous functions from $\mathbb{R}$ to $\mathbb{R}$, and suppose that $f$ and $g$ agree on the rational numbers $\mathbb{Q}$. Then $f = g$.

Note that we could prove this corollary directly using arguments based on limits (however in topology the usual notion of 'limit' does not exist).

## 4.2 Connectedness properties

### Definition 4.6

- A topological space $(X, \tau)$ is *disconnected* if there are disjoint non-empty open sets $O_1, O_2 \in \tau$ such that $X = O_1 \cup O_2$. Otherwise, $(X, \tau)$ is said to be *connected*.

  Equivalently, $(X, \tau)$ is connected if the only open sets $O \in \tau$ that are also closed are precisely $\emptyset$ and $X$.

- If $(X, \tau)$ is disconnected, then the maximal subsets $Y$ of $X$ for which $Y$ is connected (under the induced subspace topology) are referred to as the *connected components* of $X$.

  The connected components partition $X$.

**Exercise:** Show that $(X, \tau)$ is connected if and only if every continuous function $F : X \to \{0, 1\}$ is constant (here the space $\{0, 1\}$ has the discrete topology).

**Example 4.7** $\mathbb{R} \setminus \{0\}$ is disconnected but $\mathbb{R}^2 \setminus \{(0, 0)\}$ is connected. $\mathbb{R} \setminus \{0\}$ has two connected components, $(-\infty, 0)$ and $(0, \infty)$. (The proof of the first claim is easy (why?); the second and third, we will establish later).

**Remark 4.8** If $X \times Y$ is homeomorphic to $X' \times Y$ this does not imply that $X$ and $X'$ are homeomorphic! An example to show this is provided by $X = [0,1], X' = Y = [0,1)$.

**Lemma 4.9** $(X, \tau)$ is connected if and only if for every pair of non-empty subsets $A, B$ of $X$ having union $X$, we have $\overline{A} \cap B \neq \emptyset$ or $A \cap \overline{B} \neq \emptyset$.

**Theorem 4.10** The real line $\mathbb{R}$ (with the usual topology) is connected, and the connected subsets of $\mathbb{R}$ are the intervals (i.e. $(-\infty, x)$, $(-\infty, x]$, $(x, y)$, $[x, y)$, $(x, y]$, $[x, y]$, $(y, \infty)$, $[y, \infty)$ where $x < y$).

**Theorem 4.11** If $(X, \tau)$ is a connected topological space and if $f : X \to Y$ is continuous, then $f(X)$ is connected.

**Corollary 4.12 (Intermediate value theorem)** Suppose that $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) < 0$ and $f(b) > 0$. Then $f(x) = 0$ for at least one $x \in (a, b)$.

**Proposition 4.13** A product of two (or more) spaces is connected if and only if each space in the product is connected.

**Definition 4.14 (Path-connected)**
A topological space $X$ is *path-connected* if for any two points $x, y \in X$ there exists a path from $x$ to $y$ in $X$, that is, a continuous function $f : [0,1] \to X$ with $f(0) = x$ and $f(1) = y$.

**Example 4.15** $\mathbb{R}^n$ and $S^n$, $n \geq 1$, are path-connected. $\mathbb{R}^2 \setminus \{p\}$ (for any $p \in \mathbb{R}^2$) is path-connected.

**Theorem 4.16** A path-connected space is connected.

Theorem 4.16 establishes our earlier claim (Example 4.7) that $\mathbb{R}^2 \setminus \{p\}$ (for any $p \in \mathbb{R}^2$) is connected.

**Corollary 4.17** The topological spaces $\mathbb{R}$ and $\mathbb{R}^2$ are not homeomorphic.

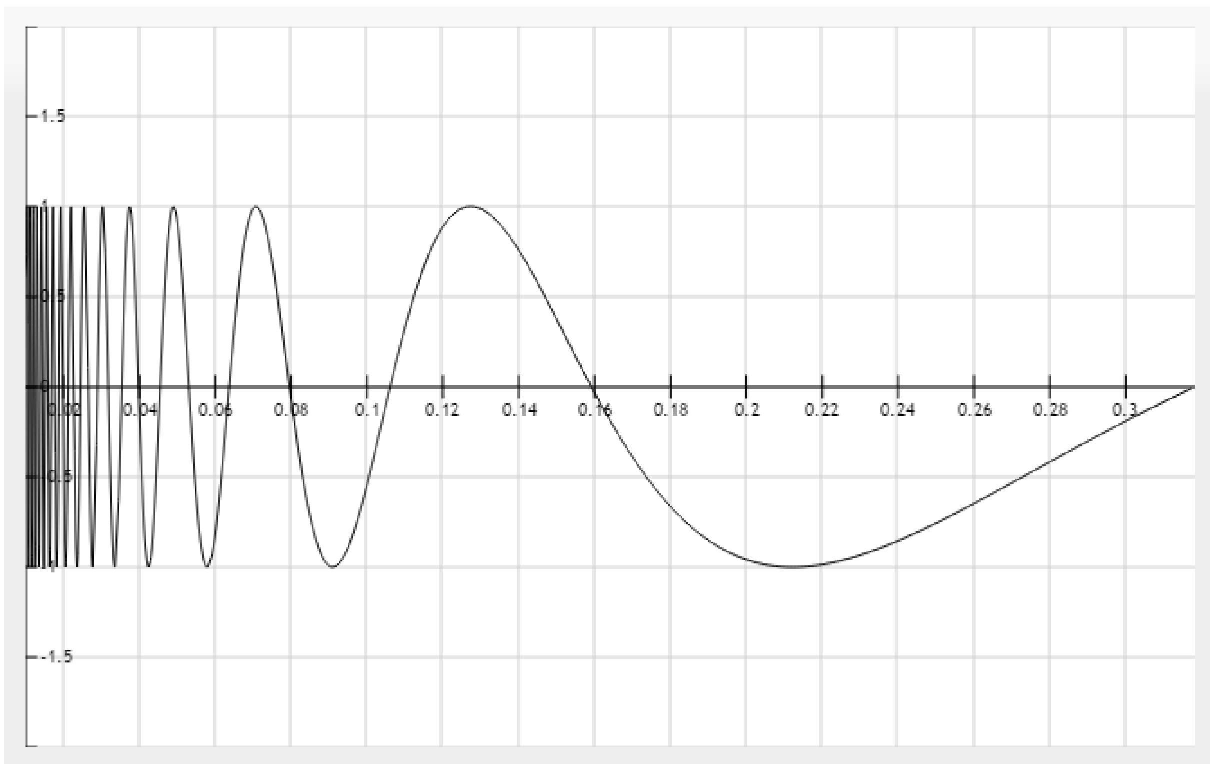
**Remark 4.18** It turns out that $\mathbb{R}^n$ and $\mathbb{R}^m$ are homeomorphic only when $n = m$, but the proof of this requires more advanced ideas than we have at this stage of the notes.

**Theorem 4.19** A connected open subset of Euclidean space is path-connected.

**Example 4.20** The topologist's sine curve

$$T = \left\{ \left( x, \sin\left(\frac{1}{x}\right) \right) \middle| x \in (0,1] \right\} \cup \{(0,0)\}$$

is connected but not path-connected.



An example of a space that is connected but not path-connected.

A further connectedness property is the following:

**Definition 4.21 (Locally path-connected topological space)**
A topological space $X$ is said to be *locally path-connected* if for each $x \in X$ and each neighbourhood $U$ of $x$ there is a path-connected neighbourhood $V$ of $x$ that is contained in $U$.

**Example 4.22** $\mathbb{R}^n$ is locally path connected.

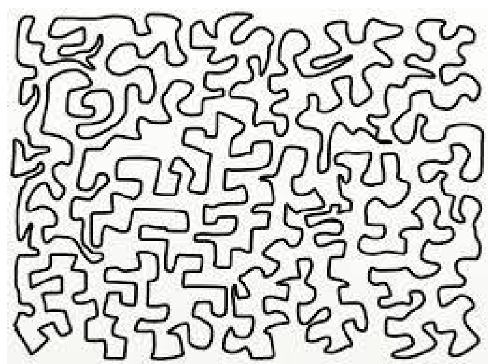
**Remark 4.23**

1. Note that a locally path-connected space need not be path-connected or even connected.

2. A connected, locally path-connected topological space is path-connected.

3. By slightly modifying the space in Example 4.20, one obtains an example of a space that is path-connected but not locally path-connected.

We finish this section by stating a version of a famous result, illustrated in the figure below. While this theorem may seem 'obvious' it is not easy to prove with the ideas we have developed so far in these notes.



An embedding of $S^1$ in $\mathbb{R}^2$

**Theorem 4.24 (Jordan Curve Theorem)**
Suppose that a subset $C$ of $\mathbb{R}^2$ is homeomorphic to $S^1$. Then $\mathbb{R}^2 \setminus C$ has exactly two connected components, one bounded (the 'inside') and the other unbounded ('the outside').

> "This is the mathematical formulation of a fact that shepherds have relied on since time immemorial!" (Laurent Siebenmann).

One can say more - in particular,

(i) each of the two components have $C$ as their boundary, and

(ii) the 'inside' and the 'outside' planar regions determined by $C$ are homeomorphic to the interior and exterior of the unit disk (Jordan–Schönflies theorem).

The Jordan Curve Theorem, and the additional point (i) extends to higher dimensions. However, (ii) famously doesn't extend to spheres in $\mathbb{R}^3$ by a construction known as the 'Alexander horned sphere'.
[VIDEO: https://www.youtube.com/watch?v=Pe2mnrLUYFU]

## 4.3 Compactness property

Compactness is an important topological property, but its definition is slightly more involved than that for Hausdorff or connectedness.

### Definition 4.25

- Given a subset $A$ of a topological space $X$, an *open cover* of $A$ is a collection $\mathcal{C}$ of open sets in $X$ whose union contains $A$.
- A *finite subcover* of $\mathcal{C}$ (for $A$) is a finite subset $F$ of $\mathcal{C}$ whose union contains $A$.
- A subset $A$ of a topological space $X$ is *compact* if every open cover of $A$ has a finite subcover.

Note that if we take $A = X$, then $X$ is (itself) compact if any collection of open sets whose union equals $X$ contains a finite subset whose union equals $X$.
Moreover, the property of $A$ being a compact subset of $X$ is equivalent to saying that $A$ with the induced subspace topology is (itself) a compact topological space.

## Example 4.26

1. $[0, 1)$ and $[0, \infty)$ are not compact.

2. The interval $[a, b]$ is compact (the Heine-Borel theorem in dimension 1).

**Theorem 4.27** A closed subspace of a compact space is compact (i.e. if $X$ is compact and $A \subset X$ is a closed set, then $A$ is compact).

**Theorem 4.28** A compact subspace of a Hausdorff space is closed.

**Theorem 4.29 (Tychonoff's theorem)**
A product of two (or more) compact spaces is compact.

Before we prove this, what, if anything, is wrong with the following proof?

The Tychonoff theorem also holds for infinite products of spaces, however the proof requires (and in fact is equivalent to!) the Axiom of Choice in set theory.

**Theorem 4.30** In a metric space, any compact subspace is closed and bounded.

[Note for a subset $A$ in a metric space $(X, d)$ to be bounded means that $A \subseteq B(x, r)$ for some $x \in X$, $r > 0$.]

**Remark 4.31** In a metric space there may exist a closed and bounded set that is not compact.

**Theorem 4.32 (Heine-Borel theorem)**
A subset $A$ of $\mathbb{R}^n$ is compact if and only if $A$ is closed and bounded (w.r.t. the Euclidean metric).

A basic property of compactness is that it is preserved by continuous functions.

**Theorem 4.33** If $f : X \to Y$ is continuous and $K \subseteq X$ is compact, then $f(K)$ is compact.

**Corollary 4.34** The sphere $S^2$ is not homeomorphic to $\mathbb{R}^2$.

**Corollary 4.35 (Extreme value theorem)**
Suppose that $K$ is a compact space and $f : K \to \mathbb{R}$ is continuous. Then $f$ is bounded and there exist $p, q \in K$ such that $f(p) = \sup_{x \in K} f(x)$ and $f(q) = \inf_{x \in K} f(x)$.

**Theorem 4.36 (Bolzano-Weierstrass property)**
An infinite subset of a compact space must have a limit point.

**Example 4.37** As an application of this last theorem, consider a billiard ball (as a point) rolling around on a frictionless billiard table. Then there is some point $p$ on the table so that for each $n \geq 1$, the ball comes within $\frac{1}{n}$ cm of $p$ at exactly midday of some day $i_n$, with $\{i_n : n \geq 1\}$ infinite.

The following result turns out to be very useful. It need not be true if $X$ is not compact (for example, consider the map $f : [0, 1) \to S^1 \subseteq \mathbb{C}$ defined by $f(x) = e^{2i\pi x}$).

**Theorem 4.38** Suppose that $f$ is a continuous one-to-one function from a compact space $X$ onto a Hausdorff space $Y$. Then $f$ is a homeomorphism.

**Remark 4.39**

1. Each of the properties 'Hausdorff', 'connected' and 'compact' is a topological property, that is, if $X$ and $Y$ are homeomorphic, then $X$ is Hausdorff if and only if $Y$ is Hausdorff, $X$ is connected if and only if $Y$ is connected, and $X$ is compact if and only if $Y$ is compact.

2. Compactness of $X$ is equivalent to the following property: If $\mathcal{C}$ is any collection of closed sets in $X$ and the intersection of every finite subcollection of sets in $\mathcal{C}$ is nonempty, then the intersection of all the sets in $\mathcal{C}$ is nonempty.

3. A space $X$ is said to be *normal* if for any two disjoint closed subsets $C, C'$ of $X$, there exist disjoint open sets $O, O'$ of $X$ with $C \subseteq O$ and

$C' \subseteq O'$. Normal spaces include all metric spaces, and all compact Hausdorff spaces (Exercise).

4. The *Urysohn metrisation theorem* states that if $X$ is a normal topological space with a countable base, then there is a metric $d$ on $X$ such that the open sets with respect to the metric $d$ are exactly the open sets of $X$.

5. A surprising result is the *Hahn-Mazurkiewicz Theorem*: A Hausdorff topological space $Y$ is the continuous image of the Cantor space $\mathcal{C}$ (described in Example 2.12) i.e. $Y = f(\mathcal{C})$ for some continuous function $f$ if and only if $Y$ is a compact, connected, locally connected metric space. In other words, there is a continuous function $f$ from $\mathcal{C}$ onto $Y$. Moreover, $f$ can be extended to $[0, 1]$.

Many important spaces are not compact but satisfy a weaker local version of compactness instead.

**Definition 4.40 (Locally compact space)**
A topological space $X$ is said to be *locally compact* if each $x \in X$ has a compact neighbourhood.

## Example 4.41

1. Every compact space is locally compact.

2. $\mathbb{R}^n$ is locally compact but not compact.

3. Every infinite discrete space is locally compact but not compact.

4. The complement of a point in a compact Hausdorff space is locally compact.

## Remark 4.42

1. A locally compact Hausdorff space $(X, \tau)$ is either compact or can be obtained from a compact Hausdorff space $(X', \tau')$ by deleting one point.

2. The space $(X', \tau')$ is called the one-point compactification of $X$. What is the one point compactification of $\mathbb{R}$? $\mathbb{R}^2$?

# 5 New spaces from old: quotient spaces and cell complexes

In this section we describe some ways of building new spaces from existing ones.

## 5.1 Quotient spaces

New topological spaces can be formed from an existing one is by 'identifying' (or 'sticking together') points of the space. This process has no real analogue in the world of metric spaces. Quotient spaces are also referred to as 'identification spaces'.

Suppose that $(X, \tau)$ is a topological space and consider any equivalence relation $\sim$ on $X$. Consider the set $X/\sim$ of equivalence classes of $\sim$ (i.e. for $x \in X$ let $[x] = \{x' \in X \mid x' \sim x\}$). Recall that the collection of equivalence classes partitions $X$. Let $p : X \to X/\sim$ be the associated projection map defined by $p(x) = [x]$.

**Definition 5.1 (Quotient space and identification map)**

Let $\sim$ be an equivalence relation on a topological space $(X, \tau)$. The set of equivalence classes $X/\sim$ becomes a topological space by declaring a subset $O$ of $X/\sim$ to be open if and only if $p^{-1}(O)$ is an open set in $X$. This space is called *quotient space* of $X$ relative $\sim$, and the associated projection map $p : X \to X/\sim$, $p(x) = [x]$, is called the *quotient* or *identification map*.

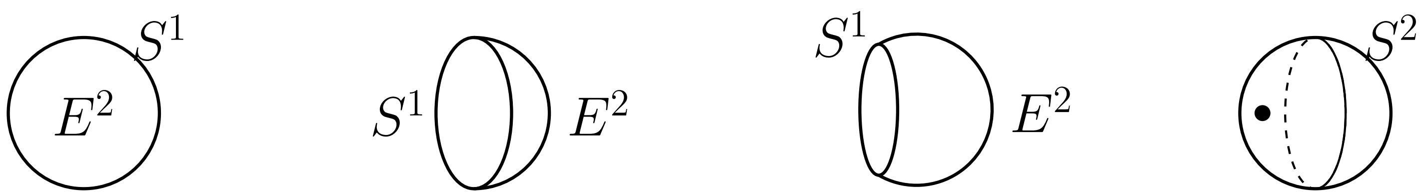
## Remark 5.2

1. Explicitly, the topology on $X/\sim$ is $\{O \subseteq X/\sim \mid p^{-1}(O) \in \tau\}$.
2. The identification map $p : X \to X/\sim$ is onto and continuous. Indeed, the topology on $X/\sim$ is the maximal one for which $p$ is continuous.

## Example 5.3

1. Consider $\mathbb{R}$ and the equivalence relation $x \sim x'$ if $x - x'$ is an integer. Then $\mathbb{R}/\sim$ is homeomorphic to $S^1$.
   $h : \mathbb{R}/\sim \to S^1 : h([x]) = e^{2\pi i x}$ is well-defined and a homeomorphism.

2. Consider $E^2$ and the equivalence relation $x \sim x'$ if $x = x'$ or $x, x' \in S^1$. Then $E^2/\sim$ is homeomorphic to the sphere $S^2$.




[A formal proof relies on Theorem 5.4 described shortly.]

3. Consider $[0, 1] \times [0, 1]$ and the equivalence relation $(x, y) \sim (x', y')$ if $(x, y) = (x', y')$ or if $x = 0, x' = 1$, and $y = y'$. Then $[0, 1] \times [0, 1]/\sim$ is homeomorphic to a cylinder.

## An important property of quotient spaces

Now suppose that we have a continuous map $f : X \to Y$. We can define an equivalence relation $\sim_f$ on $X$ by $x/\sim_f x'$ if $f(x) = f(x')$.
Suppose that $f$ is onto. A natural question is: "When is $Y$ homeomorphic to $X/\sim_f$?" Note that it is not always the case; for example, consider $X = [0, 1)$ and $Y = S^1$ and the map described earlier. However, the following result provides a positive answer.

**Theorem 5.4** If $X$ is compact, $Y$ is Hausdorff, and $f : X \to Y$ is a continuous onto map, then $X/{\sim_f}$ is homeomorphic to $Y$.

**Example 5.5** $E^2/{\sim}$ is homeomorphic to $S^2$ where $\sim$ is the equivalence relation that identifies all points in $S^1$ to a single point.

## Example 5.6 (Identification maps)

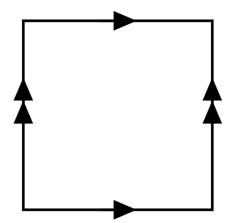
- A *Möbius strip* is the space obtained from $[0,1] \times [0,1]$ by identifying $(0, y)$ and $(1, 1-y)$ for all $y \in [0,1]$. It is similar to a cylinder, but with a 'twist'. Note that, unlike a cylinder, it only has one side and a single 'boundary edge'.

The cylinder and Möbius strip as identification spaces

It can be shown that the Möbius strip is **not** homeomorphic to the cylinder.

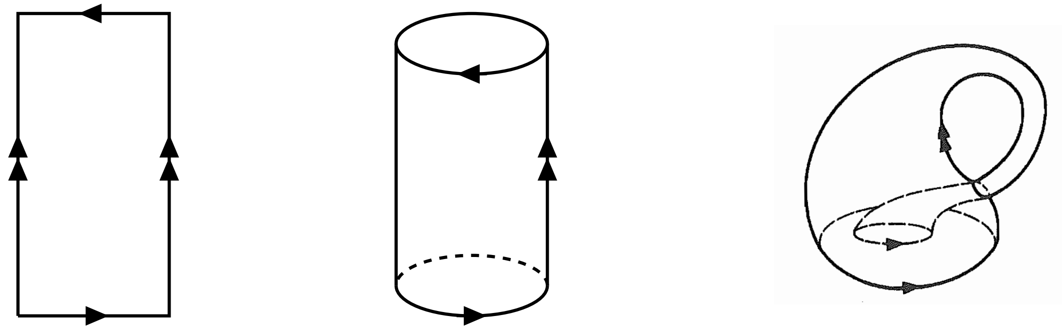
**Quiz** If you take a strip of paper and twist it twice, and glue the ends together is the resulting space homeomorphic to a cylinder or a Möbius strip or to neither?

- The *torus* ($T = S^1 \times S^1$) is the space obtained from $[0,1] \times [0,1]$ by identifying $(x,0)$ with $(x,1)$ for all $x \in [0,1]$ and $(0,y)$ with $(1,y)$ for all $y \in [0,1]$.



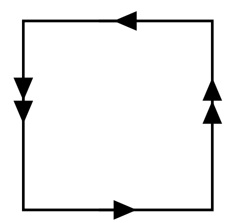
The torus as an identification space

- The *Klein bottle* ($K$) is the space obtained from $[0,1] \times [0,1]$ by identifying $(x,0)$ with $(1-x,1)$ for all $x \in [0,1]$ and $(0,y)$ with $(1,y)$ for all $y \in [0,1]$. This is illustrated in the figure below.



The Klein bottle as an identification space

- The *projective plane* $P^2$ is the topological space that corresponds to the (square) cell $[0,1] \times [0,1]$ under the identification map shown below.



The projective plane as an identification space

**Remark 5.7** Each of the above quotient spaces, except for the Möbius strip, are examples of so-called closed surfaces, see part 6 for a formal definition.

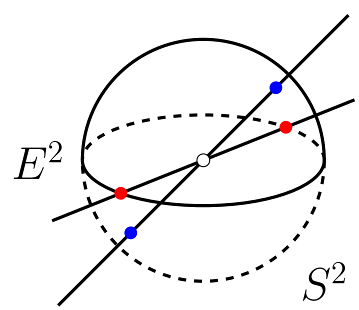
The Klein bottle $K$ and the projective plane $P^2$ cannot be embedded in $\mathbb{R}^3$ (but they can in $\mathbb{R}^4$).

**Proposition 5.8** $P^2$ is the same topological space (up to homeomorphism) as the space obtained by any one of the following three procedures:

(i) Take the circular disc $E^2$, with the equivalence $x \sim x'$ if $x = x'$ or if $x$ and $x'$ lie on the boundary of $E^2$ and $x = -x'$ (the antipodal point).

(ii) Take the sphere $S^2$ with the equivalence relation $x \sim x'$ if $x = x'$ or $x = -x'$.

(iii) Take $\mathbb{R}^3 - \{(0,0,0)\}$, with the equivalence relation $x \sim x'$ if $x$ and $x'$ lie on a line through the origin.




Different ways to describe $P^2$

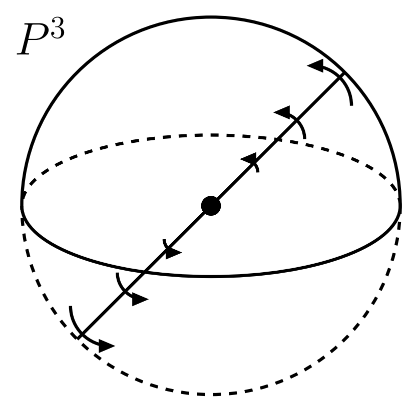
In other words, the three quotient spaces described above are all homeomorphic to each other. This space (referred to as the (real) projective plane $P^2$) is compact and connected.

Note that the space described by (iii) can be viewed as the ‘space of lines through the origin’, or more algebraically as the space of ‘1-dimensional subspaces of the 3-dimensional vector space $\mathbb{R}^3$’. In this form, replacing $\mathbb{R}$ by $\mathbb{C}$, one similarly obtains a complex projective plane.

## Remark 5.9

1. For $n \geq 2$, the space $P^n$ (called $n$–*dimensional projective space* $P^n$) is defined analogously, simply replacing $^2$ by $^n$ in (i)–(ii) above.

2. For $P^3$, there is a further way we can view this, namely as the space of rotations in $\mathbb{R}^3$; this space forms a group under composition (called $SO(3)$) and it is isomorphic to the group of $3 \times 3$ real matrices $M$ having determinant 1 and with $MM^T = I$ under matrix multiplication. This ‘topological group’ plays an important role in physics.




$P^3$ as the space of rotations in $\mathbb{R}^3$

To see how $P^3$ corresponds to the space of rotations in $\mathbb{R}^3$, recall that $P^3$ can be viewed as the three-dimensional ball $E^3$ under the equivalence relation that identifies antipodal points on the boundary sphere.

It can be shown that each rotation in $\mathbb{R}^3$ fixes some line through the origin (this follows from a more general result we will prove later, in Part 10). Thus if we take the radius of the ball $E^3$ to be $\pi$, each rotation in $\mathbb{R}^3$ can be viewed as a point $p$ in this ball, where the line of rotation is the line passing through $p$ and the origin, and the angle of clockwise rotation (facing the origin) is the distance of $p$ from the origin.

Note that this is one-to-one except for antipodal points that are identified. What does the origin correspond to?

## Spaces built up from cells

Many topological spaces can be built up by the following process by repeatedly attaching cells by their boundary to lower dimensional spaces (thus these are quotient spaces, based on cells.
First, start with the discrete space $X_0$ consisting of disjoint points (i.e. 'zero dimensional cells' $E^0 \amalg E^0 \cdots \amalg E^0$).

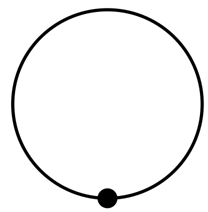
Then attach (zero or more) 1-dimensional cells $E^1$ by their boundary to $X_0$ to obtain a space $X_1$. Next, attach (zero or more) 2-dimensional cells to $X_1$ to obtain $X_2$, and so on. A space $X = X_n$ that can be constructed in this way is called a (finite dimensional) *CW complex*. Most spaces we consider from now on are of this type.

**Example 5.10** $\mathbb{R}$ can be described with $X_0 = \mathbb{Z}$ and the 1-dimensional cells of the form $[i, i+1]$ for each $i \in \mathbb{Z}$.

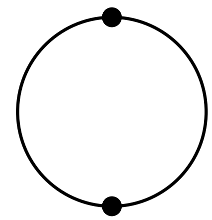
Note that if a finite number (say, $n_i$) of cells of dimension $i$ are attached for $i = 0, 1, 2, \ldots$ then the resulting space is compact. Such a space is called a *finite CW complex*, or, more informally, a *finite cell complex*.

Note that the same CW complex can be built using different numbers of cells of various dimensions, as illustrated in the figure below.

$S^1$



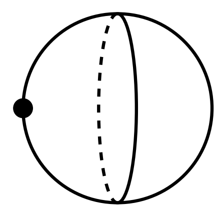
$n_0 = n_1 = 1$



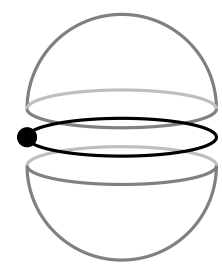
$n_0 = n_1 = 2$

To obtain $S^1$ we can start with a single $0$-cell and attach a 1-cell (so $n_0 = n_1 = 1$), or we could start with two $0$–cells and attach two 1-cells $n_0 = n_1 = 2$).

$S^2$



$n_0 = 1, n_1 = 0, n_2 = 1$



$n_0 = 1, n_1 = 1, n_2 = 2$

For $S^2$, we can start with a single $0$-cell and attach a 2-cell (so $n_0 = 1, n_1 = 0, n_2 = 1$), or we could start with a $0$-cell, attach a 1-cell, and attach two 2-cells (so $n_0 = 1, n_1 = 1, n_2 = 2$), or we could consider the cube with its $n_0 = 8$ vertices, $n_1 = 12$ edges and $n_2 = 6$ faces.

Associated with **any** topological space $X$ is an important integer-valued invariant, called the *Euler characteristic* of $X$, and written $\chi(X)$.
When $X$ is a finite cell complex then $\chi(X)$ has a simple description — it equals the number of cells of even dimension minus the number of cells of odd dimension in any cellular construction of $X$. Thus,

$$\chi(X) = n_0 - n_1 + n_2 - n_3 + \cdots .$$

It might seem that this is not well-defined, since we saw above that $X$ does not uniquely determine the $n_i$ values in the cellular description. However, $\chi(X)$ is invariant to such choices, as we now illustrate.

**Example 5.11** The Euler characteristic of $S^1$ and $S^2$ is

$$\chi(S^1) =$$

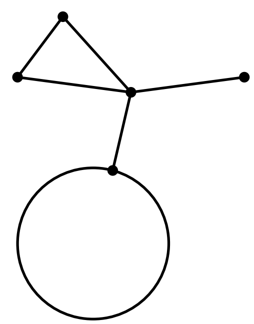
$$\chi(S^2) =$$

## Remark 5.12

1. It can also be shown that $\chi$ is a topological invariant (i.e. if $X$ and $Y$ are homeomorphic, then $\chi(X) = \chi(Y)$.
2. If you took Math320, the Euler characteristic may remind you of the Möbius function in poset theory - in fact there is a deep connection between these two (due to J.C. Rota).
3. The Euler characteristic furthermore has many other nice properties. For example, it can be shown that: $\chi(X \times Y) = \chi(X) \cdot \chi(Y)$.

**Theorem 5.13 (Euler's formula)**
If a connected graph in the plane $\mathbb{R}^2$ or on $S^2$ has $P$ vertices (points), $L$ edges (lines) no two of which cross and $F$ faces (regions bounded by edges including the unbounded region in case of $\mathbb{R}^2$), then

$$P - L + F = 2.$$



$P = 5$
$L = 6$
$F = 3$

Example of Euler's formula

**Exercise 5.14**

1. Calculate the Euler characteristic of the torus and the Klein bottle by considering a cell complex description of these spaces.

2. Show that $\chi(P^2) = 1$.

3. Show that $\chi(S^k)$ is 0 if $k$ is odd, and 2 if $k$ is even.